Week 13 – Interview preparation – Angular Track

1. What is Microsoft Azure?

Answer:
Azure is Microsoft's cloud computing platform that provides services like computing, storage, networking, databases, and more over the internet.

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2. What are the types of cloud computing services offered by Azure?

Answer:

- IaaS (Infrastructure as a Service): VMs, storage, networks
- PaaS (Platform as a Service): App Services, Azure SQL
- SaaS (Software as a Service): Microsoft 365, Dynamics 365

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3. What are the key benefits of using Azure?

Answer:

- Scalability and flexibility
- Pay-as-you-go pricing
- Global data center coverage
- High availability and disaster recovery
- Security and compliance

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4. What is the Azure Portal?

Answer:
A web-based interface to manage, configure, and monitor Azure resources.

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5. What is an Azure Subscription?

Answer:
A logical container for billing, access control, and provisioning of resources in Azure.

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6. What is an Azure Resource Group?

Answer:
A container that holds related Azure resources for easier management, deployment, and monitoring.

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7. What are Azure Regions and Availability Zones?

Answer:

- Region: A geographical location with one or more Azure data centers
- Availability Zone: Physically separate data centers within a region for high availability

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8. What is Azure Resource Manager (ARM)?

Answer:
ARM is the deployment and management service that allows you to manage Azure resources using templates, role-based access control, and tagging.

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9. What is a Tag in Azure?

Answer:
A key-value pair used to organize, categorize, and track Azure resources for management and billing purposes.

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10. What is the difference between IaaS, PaaS, and SaaS in Azure?

Answer:

- IaaS: Provides infrastructure (VMs, storage, networks)
- PaaS: Provides platform (runtime, app hosting, databases)
- SaaS: Provides software applications ready to use

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11. What is an Azure Virtual Machine (VM)?

Answer:
A VM is an on-demand, scalable computing resource running Windows or Linux.

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12. What is the difference between VM sizes in Azure?

Answer:
VM sizes define CPU, memory, storage, and network throughput for specific workloads (e.g., B-series for burstable workloads, D-series for general-purpose).

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13. What is Azure App Service?

Answer:
A PaaS offering to host web apps, REST APIs, and mobile backends without managing infrastructure.

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14. What is Azure Functions?

Answer:
A serverless compute service that runs event-driven code without the need to manage servers.

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15. What is a Virtual Machine Scale Set (VMSS)?

Answer:
A VMSS automatically deploys and manages a set of identical VMs with auto-scaling and high availability.

🔹 Azure Networking (16–20)

16. What is an Azure Virtual Network (VNet)?

Answer:
A private network in Azure that allows secure communication between Azure resources and on-premises networks.

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17. What are Subnets in Azure?

Answer:
Subnets divide a VNet into smaller IP address ranges, helping organize and secure resources.

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18. What is an Azure Network Security Group (NSG)?

Answer:
An NSG contains inbound and outbound traffic rules to allow or deny network traffic to resources.

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19. What is an Azure Load Balancer?

Answer:
A service that distributes incoming traffic across multiple resources to provide high availability.

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20. What is an Azure VPN Gateway?

Answer:
A gateway that securely connects on-premises networks to Azure VNets via encrypted VPN tunnels.

21. What are the types of data services available in Azure?

Answer:

Azure SQL Database: Managed relational database

Cosmos DB: Globally distributed NoSQL database

Azure Blob Storage: Object storage for unstructured data

Azure Table Storage: NoSQL key-value store

Azure Data Lake Storage: Big data analytics storage

Azure Database for MySQL/PostgreSQL/MariaDB: Managed open-source databases

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22. What is Azure Blob Storage used for?

Answer:
Blob storage is used to store unstructured data like text, images, videos, and backups, accessible via REST APIs.

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23. What is Azure Cosmos DB?

Answer:
Cosmos DB is a fully managed, globally distributed NoSQL database that supports multiple APIs (SQL, MongoDB, Cassandra, Table, Gremlin) with low-latency access.

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24. What is the difference between Azure SQL Database and Azure SQL Managed Instance?

Answer:

SQL Database: Platform-as-a-Service (PaaS) for single databases or elastic pools

SQL Managed Instance: PaaS with near 100% compatibility with on-prem SQL Server and instance-level features

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25. What is Azure Table Storage?

Answer:
A NoSQL key-value store for semi-structured data, suitable for storing structured metadata.

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26. What is the difference between Azure Storage Account types?

Answer:

General Purpose v2: Recommended, supports blobs, files, queues, tables

Blob Storage Account: Optimized for storing block and append blobs

FileStorage: For premium file shares

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27. What is the difference between Hot, Cool, and Archive tiers in Azure Blob Storage?

Answer:

Hot: Frequently accessed data

Cool: Infrequently accessed data

Archive: Rarely accessed, long-term storage, lowest cost

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28. What is Azure Data Lake Storage (ADLS)?

Answer:
A scalable, secure data lake for big data analytics, optimized for analytics workloads with hierarchical namespace support.

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29. How do you secure data in Azure Storage?

Answer:

Azure Active Directory integration

Shared Access Signatures (SAS)

Encryption at rest and in transit

Role-Based Access Control (RBAC)

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30. What is Azure SQL Elastic Pool?

Answer:
A resource-sharing model for multiple databases where performance resources are shared and scaled efficiently.

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31. What is Azure Service Bus?

Answer:
A fully managed message broker for enterprise messaging, supporting queues, topics, and subscriptions for asynchronous communication.

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32. What is the difference between a queue and a topic in Azure Service Bus?

Answer:

Queue: One-to-one messaging, first-in-first-out delivery

Topic: One-to-many messaging using subscriptions, allowing multiple receivers

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33. What is Azure Event Grid?

Answer:
A fully managed event routing service for event-driven architectures, enabling serverless applications and reactive workflows.

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34. What is Azure Event Hub?

Answer:
A big data streaming platform and event ingestion service capable of processing millions of events per second.

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35. How do Azure Service Bus and Event Hub differ?

Answer:

Service Bus: Enterprise messaging, ordered delivery, guaranteed delivery

Event Hub: High-throughput streaming for telemetry and logs, usually for analytics pipelines

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36. What is Azure Active Directory (AAD)?

Answer:
AAD is Microsoft's cloud-based identity and access management service used to authenticate and authorize users, applications, and devices.

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37. What is the difference between Azure AD and on-premises Active Directory?

Answer:

On-prem AD: Domain controller for local networks

Azure AD: Cloud-based identity management, supports SaaS apps and device access

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38. What are Azure AD tenants?

Answer:
A dedicated, isolated instance of Azure AD that an organization owns and manages, used to manage users, groups, and apps.

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39. What is the purpose of Azure AD Conditional Access?

Answer:
It enforces policies for controlling user access based on conditions such as location, device compliance, or risk level.

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40. What is Multi-Factor Authentication (MFA) in Azure AD?

Answer:
MFA adds an extra layer of security by requiring a second verification method (e.g., SMS, authenticator app, phone call) in addition to a password.

41. What is Azure DevOps?

Answer:
Azure DevOps is a Microsoft platform providing DevOps services for planning, developing, testing, and deploying software. It includes tools for version control, CI/CD, project tracking, and collaboration.

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42. What are the main services provided by Azure DevOps?

Answer:

Azure Repos: Version control

Azure Boards: Agile project management

Azure Pipelines: CI/CD automation

Azure Test Plans: Manual and automated testing

Azure Artifacts: Package management

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43. What are the benefits of using Azure DevOps?

Answer:

End-to-end DevOps workflow

Integration with Git and GitHub

Supports CI/CD pipelines

Agile project management tools

Cloud and on-premise support

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44. What is the difference between Azure DevOps Services and Azure DevOps Server?

Answer:

Azure DevOps Services: Cloud-hosted version of Azure DevOps

Azure DevOps Server: On-premises deployment of Azure DevOps

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45. How does Azure DevOps support Agile methodology?

Answer:
Through Azure Boards for work item tracking, sprint planning, backlogs, Kanban boards, and customizable dashboards.

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46. What is Azure Repos?

Answer:
Azure Repos provides Git repositories or Team Foundation Version Control (TFVC) for source code versioning.

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47. What is the difference between Git and TFVC in Azure Repos?

Answer:

Git: Distributed version control, local commits, branching support

TFVC: Centralized version control, single repository, server-based changes

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48. How do you create a repository in Azure Repos?

Answer:
From Azure DevOps Portal → Repos → New Repository → Choose Git or TFVC → Create.

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49. How do you clone an Azure Repos repository locally?

Answer:

git clone <repo_url>

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50. How do you create a pull request in Azure Repos?

Answer:

Push your feature branch

Navigate to Azure Repos → Pull Requests → New Pull Request

Select source and target branch, add reviewers, and create PR

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51. What is Azure Boards?

Answer:
Azure Boards is a work tracking system for planning, tracking, and discussing work across teams using work items, backlogs, boards, and sprints.

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52. What are the main work items in Azure Boards?

Answer:

Epic: High-level initiative

Feature: A smaller part of an epic

User Story / Requirement: Functional work unit

Task / Bug: Specific work or issue

53. What is the difference between a backlog and a board in Azure Boards?

Answer:

Backlog: Lists all work items, prioritized for planning

Board: Visual Kanban board for managing workflow stages

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54. What is a sprint in Azure Boards?

Answer:
A sprint is a time-boxed iteration where a team commits to complete a set of backlog items.

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55. How do you assign work items to team members in Azure Boards?

Answer:
Open the work item → Set the Assigned To field → Save.

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56. What is Azure Pipelines?

Answer:
Azure Pipelines is a CI/CD service that automates building, testing, and deploying applications to any environment.

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57. What are the main components of an Azure Pipeline?

Answer:

Pipeline: Defines CI/CD workflow

Agent: Executes tasks

Stages: Logical phases of the pipeline

Jobs: Units of work in stages

Tasks/Steps: Individual actions like build, test, or deploy

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58. What is the difference between build and release pipelines?

Answer:

Build Pipeline: Compiles code, runs tests, produces artifacts

Release Pipeline: Deploys artifacts to different environments

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59. What is YAML in Azure Pipelines?

Answer:
YAML is a human-readable file format used to define pipelines as code, allowing version control and repeatable builds/deployments.

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60. How do you trigger a pipeline automatically on code changes?

Answer:
Set up Continuous Integration (CI) triggers in the pipeline YAML file or through pipeline settings to start builds automatically on commits or PR merges.

61. What are the differences between classic and YAML pipelines in Azure DevOps?

Answer:

Classic Pipelines: Configured via GUI, easier for beginners, less version control-friendly

YAML Pipelines: Defined as code in a .yaml file, stored in repo, supports CI/CD as code, better for automation and reproducibility

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62. What are pipeline templates in Azure Pipelines?

Answer:
Pipeline templates allow you to define reusable YAML snippets for stages, jobs, or steps that can be included across multiple pipelines.

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63. What are stages in Azure Pipelines?

Answer:
Stages are logical sections of a pipeline that group jobs for deployment or testing. Each stage can have its own jobs, conditions, and approvals.

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64. How do you implement environment-specific deployments in YAML pipelines?

Answer:
Use variables, variable groups, or environments to define settings for each environment (e.g., Dev, QA, Prod).

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65. What is the difference between an agent pool and an agent in Azure Pipelines?

Answer:

Agent: A machine that executes pipeline jobs

Agent Pool: A collection of agents from which jobs are picked for execution

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66. What is the purpose of a pipeline artifact?

Answer:
Artifacts are files or packages produced by a build pipeline that can be used in release pipelines for deployment.

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67. What are triggers in Azure Pipelines?

Answer:
Triggers define when a pipeline runs automatically, such as CI triggers on code commits, PR triggers, or scheduled triggers.

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68. How do you implement approvals and checks in a pipeline?

Answer:
Set environment approvals in YAML or classic pipelines to require manual verification before deployment to a stage or environment.

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69. What are conditions in Azure Pipelines?

Answer:
Conditions determine whether a job or step runs based on criteria like branch, previous job success, or variable values.

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70. How can you implement parallel jobs in a pipeline?

Answer:
Use multiple jobs in the same stage or define matrix strategies to run tasks concurrently across different configurations.

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71. What is an Azure Release Pipeline?

Answer:
A release pipeline automates deployment of artifacts from build pipelines to various environments such as Dev, QA, and Production.

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72. What is a deployment group in Azure DevOps?

Answer:
A deployment group is a collection of target machines (servers) with agents installed, used for deploying applications in a controlled manner.

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73. What are the different deployment strategies supported in Azure Pipelines?

Answer:

Blue-Green Deployment: Two identical environments, traffic shifted to new version

Canary Deployment: Gradually roll out changes to a small subset of users

Rolling Deployment: Deploy sequentially across multiple instances

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74. How do you define pipeline approvals for release pipelines?

Answer:
Set pre-deployment approvals or post-deployment approvals for stages, requiring designated users or groups to approve before proceeding.

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75. What is the difference between artifacts and releases in Azure Pipelines?

Answer:

Artifacts: Build outputs (code packages, binaries)

Release: The process of deploying artifacts to environments using a release pipeline

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76. What is Azure Test Plans?

Answer:
Azure Test Plans is a service in Azure DevOps for managing, planning, executing, and tracking manual and exploratory testing.

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77. What are test suites in Azure Test Plans?

Answer:
Test suites are collections of test cases grouped logically for execution. Types include static, requirement-based, and query-based suites.

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78. What is a test case in Azure Test Plans?

Answer:
A test case is a single unit of testing that defines input, expected result, and steps to validate functionality.

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79. How can you run automated tests in Azure Pipelines?

Answer:
Integrate automated test scripts (unit, integration, UI) in build or release pipelines using tasks like Visual Studio Test, NUnit, Selenium, or REST APIs.

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80. How do you track test results in Azure Test Plans?

Answer:
Test results are automatically captured during execution and can be analyzed via Test Runs, Test Reports, Dashboards, and historical trend charts.